Peter Emad

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Portfolio | GitHub | LinkedIn | Codeforces

Education

Degree/Certificate	University/School	Percentage	Year
Bachelor	Faculty of Engineering Helwan University, Cairo	70.53% (Current)	2023-Present
-	Faculty of Engineering Beni-Sueif University	84.07%	2022-2023
High school diploma	Elmassria integrated language schools	84.63%	2008-2022

Courses

- **Arduino** *August 2023*
  Certificate October 2023
  I have achieved
  - Basics of programming in Arduino C
  - Sensors & Modules (LM35, Flame, Gas, Ultrasonic, etc.)
  - PCB Implementation & Testing

- **Problem Solving Level 0 in C++** *October 2023*
  Certificate January 2024
  I have accomplished
  - Data-type & IF-Conditions
  - Loops, Arrays, Strings, Functions
  - Math-Geometry

- **Problem Solving Level 1 in C++ (PIXELS)** *February 2024*
  Certificate February 2024
  I have accomplished
  - STLS1 & STLS2
  - Binary Search
  - Bitmask
  - Backtrack Algorithms
  - Graph Algorithms: PFS & DFS, Dijkstra

- **Embedded (IEEE)** *February 2024*
  Certificate August 2024
  I have mastered
  - Embedded system concepts & computer architecture
  - Programming in C & Embedded C
  - Interfacing
  - Coding Drivers (DIO, LCD, ADC, Keypad, etc.)

- **CS50 (Harvard University)** *September 2024*
  Certificate October 2024
  I have learned
  - Programming Fundamentals: C, Python, SQL, HTML, CSS, JavaScript
  - Algorithms & Data Structures: Arrays, Linked Lists, Stacks, Queues, Hash Tables,Searching & Sorting algorithms
  - Problem-Solving & Computational Thinking
  - Memory Management & Pointers
  - Web Development: Building dynamic websites with Flask
  - Software Engineering Principles: Testing, debugging, and version control with Git
  - Project Development: Completed multiple problem sets and a final project demonstrating practical coding skills
  - This was a comprehensive course that helped in expanding my understanding of computer science and programming, focusing on algorithms, memory management, and web development

- **Machine Learning (IEEE)** *January 2025*
  Certificate March 2025
  I have acknowledged
  - Python: NumPy, Pandas, TensorFlow
  - Data visualization
  - Linear Regression, Logistic Regression
  - Data visualization

- KNN & K-Means
- Neural Networks
- CNN

- **Digital IC design** *January 2025*
  Certificate March 2025
  Gained experience in
  - Logic gates, Latches, Flip Flops
  - Transistors
  - Integrated & Discrete circuits
  - Verilog

- **Internet of Things (NTI)** *July 2025*
  Certificate Aug 2025
  Gained experience in
  - Introduction to Internet of Things (IoT).
  - IoT Architectures and protocols.
  - Introduction to C programming.
  - Sensors / Actuators interfacing with PI to implement IoT.
  - Implement IoT Broker and clients.
  - Packet Tracer - Sensors and the PT Microcontroller.
  - Sensors / Actuators interfacing for IoT.
  - Fog and Cloud Computing.
  - Developing IoT applications.
  - Industrial Internet of Things (IIoT).
  - IoT platform implementation based on MQTT.
  - IoT based projects / ex. smart city, smart home, smart factory, e-health over IoT.

- **AI (ERI)** *July 2025*
  Certificate Aug 2025
  Gained experience in
  - Fundamental Understanding of AI Concepts.
  - Practical Skills in AI Development.
  - Core Knowledge of Machine Learning.
  - Familiarity in Neural Networks and Deep Learning.
  - Project Development and Deployment Skills.

Honors

- **Mentor in FEHU-CPC Community** *Former Mentor*
  2023-2024
  - I provide guidance for beginners, explain algorithms, data structures, and help learners develop problem-solving skills

Projects

- **Personal Portfolio Website** *2026*
  Next.js, React, TypeScript, Tailwind CSS GitHub
  - Designed and developed a modern responsive portfolio website using Next.js and React
  - Implemented animated UI components, smooth scrolling, and interactive sections
  - Integrated GitHub activity heatmap and dynamic project showcase
  - Used Tailwind CSS for modern UI styling and responsive design

- **Hand Gesture Media Controller** *2025*
  Python, OpenCV, MediaPipe, Pycaw GitHub
  - Developed a hand gesture recognition system to control media playback and system volume
  - Implemented real-time hand tracking using MediaPipe and OpenCV
  - Integrated Pycaw library for Windows audio control
  - Optimized performance for real-time gesture detection

- **Explainable AI Analysis Project** *2025*
  Python, Scikit-learn, SHAP, LIME GitHub
  - Built machine learning models to analyze product rating data
  - Implemented Explainable AI techniques using SHAP and LIME
  - Evaluated model predictions and visualized feature importance
  - Performed data preprocessing, model training, and evaluation

- **Terminal-Based Tic Tac Toe with AI** *2025*
  C++ GitHub
  – Implemented a terminal-based Tic Tac Toe game with AI opponent
  – Used Minimax algorithm with Alpha-Beta pruning for optimal gameplay
  – Added multiple AI difficulty levels and score tracking
  – Designed ANSI color-based UI for better user interaction

- **Traffic Light Countdown System** *2024*
  Arduino GitHub
  – Designed a traffic light system with pedestrian button and countdown timer
  – Implemented embedded control logic using Arduino C
  – Designed and tested a PCB for the system
  – Ensured safe timing logic for vehicle and pedestrian signals

Skills

- **Programming**: C/C++, Python
- **Web Development**: HTML, CSS, JavaScript, TypeScript
- **Frontend Frameworks**: React.js, Next.js
- **Styling**: Tailwind CSS
- **Problem Solving**: C++
- **Embedded Systems**: C
- **Arduino**: Arduino C
- **Unity**: C#

Certifications

- **Arduino**: Arduino Certificate
- **Problem Solving**: Problem Solving Level 1 Certificate
- **Embedded**: Embedded (IEEE) Certificate
- **IOT**: IOT (NTI) Certificate
- **Software Engineering**: cs50 (Harvard) Certificate
- **AI**: (ERI) Certificate

Achievements

- **Best Student of the Month**, Arduino Award *2024*
- **Best Code of the Week**, Arduino Award *2024*

Languages

- **Arabic**, Native language
- **English**, Very good

Hobbies

- **Problem Solving**, Competitive programming in C++
- **Math**, Pursuing math to hone critical thinking and optimization skills

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This CV was created using LaTeX by Peter Emad.